Ventilator



Category: **Medical, Healthcare & First Aid** Type: **Full SOP** Priority: **High** 2 labs

Covers these lab items: Ventilator

What it is used for

A medical ventilator supports or controls a patient's breathing by delivering a prescribed breathing pattern and oxygen concentration. It must be operated by trained healthcare professionals using a patient-specific clinical plan and continuous monitoring.

Main parts

- Ventilator unit and display
- Patient breathing circuit and tubing
- Breathing interface or endotracheal/tracheostomy connection
- Air and oxygen supply connections, as applicable
- Humidifier and compatible chamber, if prescribed
- Alarms, sensors, filters, and power backup

Before you start

1. Verify the patient's identity and the prescribed ventilation plan.
2. Only trained and authorised clinicians should prepare the ventilator and circuit.
3. Inspect the unit, breathing circuit, filters, humidifier, connectors, and power supply.
4. Check oxygen and compressed-air supplies or internal gas sources, as applicable.
5. Complete the manufacturer's self-test and circuit leak test before connecting to the patient.
6. Confirm alarm limits, battery backup, emergency ventilation equipment, and monitoring are available.

How to use

1. Assemble the circuit and patient interface using the approved infection-control procedure.
2. Select the prescribed ventilation mode and enter patient-specific settings only as authorised by the clinical team.
3. Verify settings, alarm limits, gas supply, and circuit connections before use.
4. Connect the ventilator to the patient only under the responsible clinician's direction.
5. Confirm effective ventilation using appropriate clinical observations and monitoring, including oxygenation and carbon dioxide where indicated.
6. Observe chest movement, patient comfort, airway pressure, delivered volumes, and displayed waveforms as clinically appropriate.
7. Respond promptly to alarms by assessing the patient first and checking the circuit and gas supply.

8. If ventilation fails, follow the emergency protocol and use approved backup ventilation when directed.
9. Document settings, observations, alarms, interventions, and changes according to clinical policy.
10. When ventilation is discontinued, follow the authorised weaning or disconnection plan.
11. Disconnect and dispose of or reprocess circuit components according to infection-control instructions.
12. Clean the device exterior and prepare the ventilator for the next approved use.

⚠ Safety

- Never use a ventilator without appropriate clinical training and supervision.
- Ventilator settings must be prescribed and adjusted by authorised healthcare professionals.
- Do not silence or disable alarms without an approved clinical reason and monitoring plan.
- Maintain a backup ventilation method and emergency response plan.
- Use infection-control precautions and patient-specific or appropriately reprocessed circuits and accessories.
- If a critical alarm occurs, assess the patient immediately and follow the emergency protocol.

Care & storage

- Clean and disinfect external surfaces using manufacturer-approved products.
- Replace filters and circuit components according to the instructions and infection-control policy.
- Check battery backup, alarms, sensors, and self-test results at the required intervals.
- Arrange preventive maintenance and calibration by authorised biomedical personnel.
- Remove faulty equipment from clinical use and report the defect promptly.

Common mistakes to avoid

- Using settings that do not match the authorised patient-specific plan.
- Ignoring alarms or failing to assess the patient when they sound.
- Using an incorrectly assembled, leaking, or contaminated breathing circuit.
- Failing to check backup ventilation, gas supplies, or power backup.

Disclaimer: This SOP is prepared from general information available on the internet. For detailed instructions, refer to the manufacturer's original manual / file.